

6. The table below shows the mean carbon footprint for 1 kWh of electricity generated in different ways.

Type of power station	Carbon footprint (gCO ₂ eq/kWh)
coal	921
oil	750
gas	500
solar	100
biomass	52
nuclear	30
hydroelectric	27
wind	30

Use the information in the table and your knowledge to answer the following questions.

- (a) (i) Define the term carbon footprint. [2]

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- (ii) A typical 500 MW **coal**-fired power station produces 3.5 billion kWh of electricity per year.
Calculate the carbon footprint of this power station in gCO₂eq/kWh per year.
Give your answer to 2 significant figures.

1 billion kWh = 1×10^9 kWh [4]

carbon footprint = gCO₂eq/kWh



- (iii) The carbon footprint of a nuclear power station is close to zero whilst generating electricity.

Explain why the table shows a carbon footprint of 30 gCO₂eq/kWh for power generated by nuclear power.

[2]

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- (iv) A biomass power station burns wood chips as its energy source.
State why a biomass generator is considered as being 'carbon neutral'.

[1]

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- (b) The Antarctic ice sheet traps methane. It is estimated that the mass of trapped methane is about 2.8×10^{17} kg which has a carbon dioxide equivalent of 7×10^{18} kgCO₂eq.

- (i) Explain what is meant by the term global warming potential.

[1]

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- (ii) Use the equation:

carbon dioxide equivalent = mass of a gas $\times$ global warming potential of the gas
to calculate the global warming potential of methane.

[2]

global warming potential =

END OF PAPER

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THE PERIODIC TABLE

Group

2
1

3 4 5 6 7 0

1H Hydrogen 1										4He Helium 2																											
7Li Lithium 3		9Be Beryllium 4																		11B Boron 5		12C Carbon 6		14N Nitrogen 7		16O Oxygen 8		19F Fluorine 9		20Ne Neon 10							
23Na Sodium 11		24Mg Magnesium 12																		27Al Aluminium 13		28Si Silicon 14		31P Phosphorus 15		32S Sulfur 16		35.5Cl Chlorine 17		40Ar Argon 18							
39K Potassium 19		40Ca Calcium 20		45Sc Scandium 21		48Ti Titanium 22		51V Vanadium 23		52Cr Chromium 24		55Mn Manganese 25		56Fe Iron 26		59Co Cobalt 27		59Ni Nickel 28		63.5Cu Copper 29		65Zn Zinc 30		70Ga Gallium 31		73Ge Germanium 32		75As Arsenic 33		79Se Selenium 34		80Br Bromine 35		84Kr Krypton 36			
86Rb Rubidium 37		88Sr Strontium 38		89Y Yttrium 39		91Zr Zirconium 40		93Nb Niobium 41		96Mo Molybdenum 42		99Tc Technetium 43		101Ru Ruthenium 44		103Rh Rhodium 45		106Pd Palladium 46		108Ag Silver 47		112Cd Cadmium 48		115In Indium 49		119Sn Tin 50		122Sb Antimony 51		128Te Tellurium 52		127I Iodine 53		131Xe Xenon 54			
133Cs Caesium 55		137Ba Barium 56		139La Lanthanum 57		179Hf Hafnium 72		181Ta Tantalum 73		184W Tungsten 74		186Re Rhenium 75		190Os Osmium 76		192Ir Iridium 77		195Pt Platinum 78		197Au Gold 79		201Hg Mercury 80		204Tl Thallium 81		207Pb Lead 82		209Bi Bismuth 83		210Po Polonium 84		210At Astatine 85		222Rn Radon 86			
223Fr Francium 87		226Ra Radium 88		227Ac Actinium 89																																	

Key

relative atomic mass

A_r	Symbol	Name	Z
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